

## MATH 261: Exam 1 Practice Problems

1. Let  $\mathbf{u} = 1\mathbf{i} + 1\mathbf{j} + 1\mathbf{k}$  and  $\mathbf{v} = 0\mathbf{i} + -1\mathbf{j} + 2\mathbf{k}$ . Compute  $3\mathbf{u} + \mathbf{v}$ .
2. Let  $\mathbf{v} = 2\mathbf{i} + 1\mathbf{j} + -2\mathbf{k}$ . Find a unit vector that points in the same direction as  $\mathbf{v}$ .
3. Let  $\mathbf{u} = \langle 1, -1, 1 \rangle$  and  $\mathbf{v} = \langle 2, 2, 0 \rangle$ . Compute the dot product  $\mathbf{u} \cdot \mathbf{v}$  and the cross product  $\mathbf{u} \times \mathbf{v}$ .
4. Suppose that  $\mathbf{u}$  and  $\mathbf{v}$  are two nonzero vectors. What would the equation  $\mathbf{u} \cdot \mathbf{v} = 0$  tell you about  $\mathbf{u}$  and  $\mathbf{v}$  geometrically? What would the equation  $\mathbf{u} \times \mathbf{v} = 0$  tell you about  $\mathbf{u}$  and  $\mathbf{v}$  geometrically?
5. If  $\mathbf{u} \cdot \mathbf{v} = 2$ , what must  $3\mathbf{u} \cdot 2\mathbf{v} + \mathbf{v} \cdot 4\mathbf{u}$  equal?
6. If  $\mathbf{u} \times \mathbf{v} = 2\mathbf{j} - \mathbf{k}$ , what must  $2\mathbf{v} \times \mathbf{u}$  equal?
7. Find the parametric equations of the line that goes through the point  $(3, 1, -1)$  and is parallel to the vector  $\langle 2, 4, 1 \rangle$ . At what point or points does this line intersect the plane  $x - y + 2z = 0$ , if any?
8. Find the parametric equations of the line that goes through the points  $(1, 2, 2)$  and  $(3, 3, 0)$ . At what point or points, if any, does this line intersect the plane  $x + z = 1$ , if any?
9. At which point or points, if any, does the line through the points  $(2, 3, 1)$  and  $(3, 4, 1)$  intersect the line which contains the point  $(0, 2, 0)$  and is parallel to the vector  $\langle 1, 0, 1 \rangle$ ?
10. Find an equation for the plane which contains the point  $(2, 3, 5)$  and whose normal line is in the direction of  $\langle 1, 1, -1 \rangle$ .
11. Find an equation for the plane which contains the points  $(1, 1, 2)$ ,  $(4, 5, -2)$ , and  $(0, 0, 1)$ .
12. Find an equation for the plane which contains the points  $(2, 1, 0)$ ,  $(1, 0, 2)$ , and  $(0, -1, -4)$ .
13. Find a parametric equation for the line on which the planes  $x + y - z = 0$  and  $x + z = 3$  meet.
14. Find a parametric equation for the line on which the planes  $x + 2y + z = 0$  and  $x - y + 2z = 4$  meet.
15. Find a parametric equation for the circle  $x^2 + y^2 = 1$ .
16. Find a parametric equation for the circle  $x^2 + y^2 = 16$ .
17. Find a parametric equation for the ellipse  $x^2 + \frac{y^2}{81} = 1$ .
18. Find a parametric equation for the ellipse  $\frac{x^2}{25} + \frac{y^2}{9} = 100$ .
19. Find a parametric equation for the triangle with vertices at  $(0, 0)$ ,  $(2, 0)$ , and  $(0, 5)$ .

20. Consider the curve parameterized by  $\mathbf{r}(t) = (t^3, e^t + 1)$  for  $0 \leq t \leq 1$ .
- What are the starting and ending points of this curve?
  - If this curve actually represented the position of an object, what would be the velocity vector of the object?
  - Compute an expression for the unit tangent vector,  $\mathbf{T}(t)$ .
21. Suppose  $f(x, y) = \sin(x^2y)$ . Compute the partial derivatives  $f_x(x, y)$  and  $f_y(x, y)$ .
22. Suppose  $f(x, y) = x \tan(x + y)$ . Compute the partial derivatives  $f_x(x, y)$  and  $f_y(x, y)$ .
23. Suppose  $f(x, y) = \ln(100 + x^2 + y^4)$ . Compute the partial derivatives  $f_x(x, y)$  and  $f_y(x, y)$ .
24. Suppose  $f(x, y) = x^3y + y^2x^2 + 2x - y + 12$ . Compute the partial derivatives  $f_x(x, y)$ ,  $f_y(x, y)$ ,  $f_{xx}(x, y)$ ,  $f_{yy}(x, y)$  and  $f_{xy}(x, y)$ .
25. Suppose  $f(x, y, z) = x + x^2y^2 + x^2y^2z^2$ . Compute the partial derivatives  $f_x(x, y, z)$ ,  $f_y(x, y, z)$ , and  $f_z(x, y, z)$ .
26. Suppose  $\nabla f = 2\mathbf{i} - 3\mathbf{j} + \mathbf{k}$  at  $(0, 0, 0)$ . What is the directional derivative of this function at the origin,  $D_{\mathbf{u}}f(0, 0, 0)$ , where  $\mathbf{u}$  is a vector which points in the direction of  $\mathbf{u} = \langle 1, 1, -1 \rangle$ ?
27. Is the function  $f(x, y) = x^2y^2 + x^4 + e^y$  increasing, decreasing or instantaneously constant at the point  $(-1, 0)$  in the direction  $\langle 1, 1 \rangle$ ?
28. Is the function  $f(x, y, z) = ze^{x^2y}$  increasing, decreasing or instantaneously constant at the point  $(1, 1, 1)$  in the direction  $\langle 3, -1, -2 \rangle$ ?
29. Suppose  $g(x, y) = ye^x$ .
- Compute the gradient  $\nabla g(x, y)$ .
  - What is the direction of greatest increase for this function at  $(0, 0)$  (expressed as a unit vector)?
  - Compute the directional derivative  $D_{\mathbf{u}}f(x, y)$  at the point  $(0, 0)$ , where  $\mathbf{u}$  points in the direction of  $\langle 1, -2 \rangle$ .
  - Compute the directional derivative  $D_{\mathbf{u}}f(x, y)$  at the point  $(0, 0)$ , where  $\mathbf{u}$  points in the direction of  $\langle 2, -4 \rangle$ .
  - Compute the directional derivative  $D_{\mathbf{u}}f(x, y)$  at the point  $(0, 0)$ , where  $\mathbf{u}$  points in the direction of  $\langle 5, 0 \rangle$ .
30. Suppose  $g(x, y) = e^{-x^2y^2}$ .
- Compute the gradient  $\nabla g(x, y)$ .
  - What is the direction of greatest increase for this function at  $(1, -1)$  (expressed as a unit vector)?
  - Compute the directional derivative  $D_{\mathbf{u}}f(x, y)$  at the point  $(1, -1)$ , where  $\mathbf{u}$  points in the direction of  $\langle 2, 1 \rangle$ .
31. Suppose  $h(x, y, z) = x^2y + y^2x + z^3$ .
- Compute the gradient  $\nabla h(x, y, z)$ .

- b) What is the direction of greatest increase for this function at  $(1, 1, 1)$  (expressed as a unit vector)?
- c) Compute the directional derivative  $D_{\mathbf{u}}f(x, y)$  at the point  $(1, 1, 1)$ , where  $\mathbf{u}$  points in the direction of  $\langle 1, 2, 1 \rangle$ .
32. Find an equation of tangent plane to the surface  $x^2 + y^2 + z^2 = 3$  at the point  $(1, 1, 1)$ .
33. Find an equation of tangent plane to the surface  $\cos(\pi x) - x^2y + e^{xz} + yz = 4$  at the point  $(0, 1, 2)$ .
34. Find an equation of tangent plane to the surface  $z = \ln(x^2 + y^2)$  at the point  $(1, 0, 0)$ .
35. Find an equation of tangent plane to the surface  $z = \sqrt{y - x}$  at the point  $(1, 2, 1)$ .
36. Is  $(1, 1)$  a critical point of the function  $f(x, y) = x^4 - y^2x^2 + x^2 - y^2$ ? If so, classify it as a local maximum, minimum, or saddle point. If not, explain why not.
37. Suppose  $f(x, y)$  is a function so that  $f_x(1, 2) = 0$ ,  $f_y(1, 2) = 0$ , and suppose that  $f_{xx}(1, 2) = 3$ ,  $f_{yy}(1, 2) = 2$ , and  $f_{xy}(1, 2) = 1$ . Is  $(1, 2)$  a local maximum, local minimum, or saddle point for this function?
38. Suppose  $f(x, y)$  is a function so that  $f_x(1, 2) = 0$ ,  $f_y(1, 2) = 0$ , and suppose that  $f_{xx}(1, 2) = 5$ ,  $f_{yy}(1, 2) = 3$ , and  $f_{xy}(1, 2) = 4$ . Is  $(1, 2)$  a local maximum, local minimum, or saddle point for this function?
39. Consider a function  $f(x, y) = xy - x^2 - y^2 - 2x - 2y + 4$ ,
- Find critical point(s).
  - Use the second derivative test to determine if each critical point corresponds to a local maximum, local minimum or saddle point? If it corresponds to a local maximum or minimum, find its value.
40. Consider a function  $f(x, y) = -3x^2 + 3y^2 - 2y^3 + 6xy$ ,
- Find critical point(s).
  - Use the second derivative test to determine if each critical point corresponds to a local maximum, local minimum or saddle point? If it corresponds to a local maximum or minimum, find its value.

41. Use the Chain Rule to evaluate  $\frac{df}{dt}$  for the following function.

$$f(x, y) = x^2y + 3xy^4, \quad x(t) = \sin(2t), \quad y(t) = \cos(t)$$

42. Use the Chain Rule to evaluate  $\frac{df}{dt}$  for the following function.

$$f(x, y) = e^{x^2-y}, \quad x = \sin(4t), \quad y = t^3 - 9$$

43. Use the Chain Rule to evaluate  $\frac{df}{dt}$  for the following function.

$$f(x, y, z) = z - \sin(xy), \quad x = t, \quad y = \ln t, \quad z = e^{t-1}$$

44. Use the Chain Rule to evaluate  $\frac{df}{dt}$  for the following function.

$$f(x, y, z) = x^4 - 4xy^2 + z^3, \quad x = \sqrt{t}, \quad y = e^{2t}, \quad z = \frac{1}{t}$$

45. Use the Chain Rule to evaluate  $\frac{\partial z}{\partial s}$  and  $\frac{\partial z}{\partial t}$  for the following function.

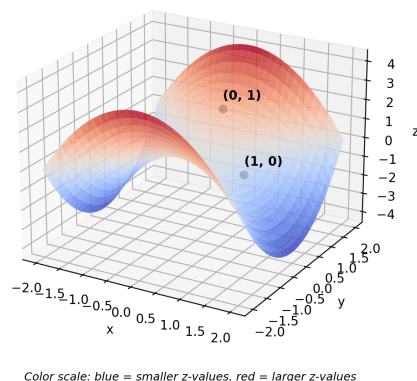
$$z = x^2y + x, \quad x = s^2 + 3t, \quad y = 2s - t$$

46. Use the Chain Rule to evaluate  $\frac{\partial w}{\partial s}$  and  $\frac{\partial w}{\partial t}$  for the following function.

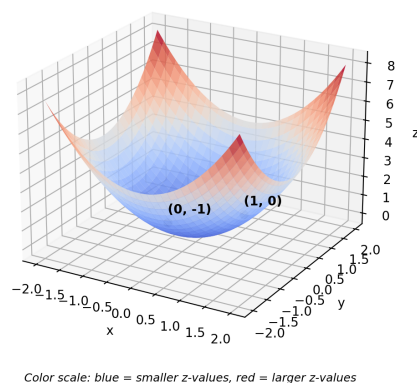
$$w = xy + z^2, \quad \text{where } x = t^2e^s, \quad y = t \cos(s)$$

47. (Conceptual / Visualization)

The surface  $z = f(x, y) = y^2 - x^2$  is shown below, with two points marked on the surface.



- (a) At the point  $(1, 0)$ , holding  $y = 0$  fixed, is the cross-section of the surface increasing or decreasing as  $x$  increases? What does this tell you about the sign of  $\frac{\partial f}{\partial x}$  at  $(1, 0)$ ?
- (b) At the point  $(0, 1)$ , use the graph to determine the sign of  $\frac{\partial f}{\partial y}$ .
- (c) Based on the shape of the surface, classify the critical point at the origin as a local maximum, local minimum, or saddle point. Explain your reasoning using the graph alone (no computation required)
48. (Conceptual / Visualization)
- The surface  $z = f(x, y) = x^2 + y^2$  is shown below, with two points marked on the surface.



- At the point  $(1, 0)$ , holding  $y = 0$  fixed, is the cross-section increasing or decreasing as  $x$  increases? What does this tell you about the sign of  $\frac{\partial f}{\partial x}$  at  $(1, 0)$ ?
- At the point  $(0, -1)$ , use the graph to determine the sign of  $\frac{\partial f}{\partial y}$ .
- Based on the shape of the surface, classify the critical point at the origin as a local maximum, local minimum, or saddle point. Explain your reasoning using the graph alone (no computation required)

49. (Conceptual / Visualization)

Consider the surface  $z = f(x, y) = 18 - x^2 - 3y^2$ , shown in Figure (a), with the point  $(2, 1, 11)$  marked.

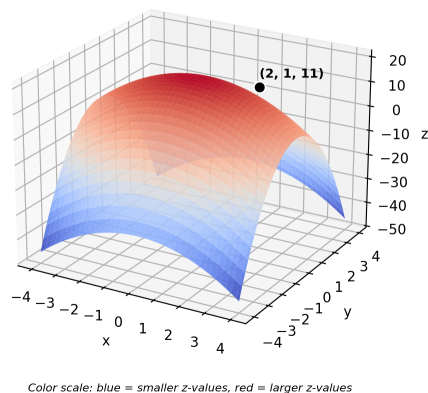


Figure (a)

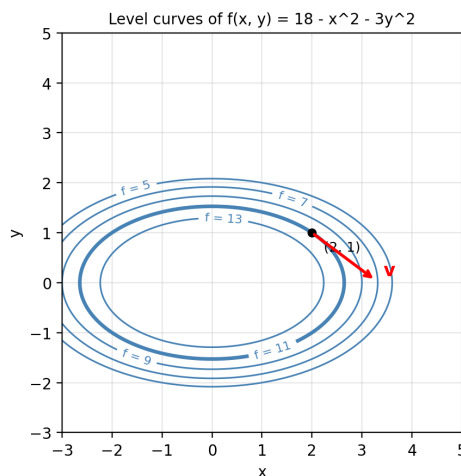


Figure (b)

- Compute  $\nabla f$  at  $(2, 1)$ .
- The level curve of  $f$  through  $(2, 1)$  (i.e.,  $f(x, y) = 11$ ) is shown in Figure (b), along with two neighboring level curves on either side for context. On the graph, draw the gradient vector  $\nabla f$  at  $(2, 1)$ , in the correct direction (you do not need to scale it precisely). Briefly explain how you knew which way it should point.

- (c) The red arrow in Figure (b) represents the direction vector  $\mathbf{v} = \langle 4, -3 \rangle$ . Compute the directional derivative of  $f$  at  $(2, 1)$  in that direction, and determine whether  $f$  is increasing or decreasing. Use the graph to confirm your conclusion.

50. (Conceptual / Visualization)

Consider the surface  $z = f(x, y) = x^2 - y^2$ , shown in Figure (a), with the point  $(1, 2, -3)$  marked.

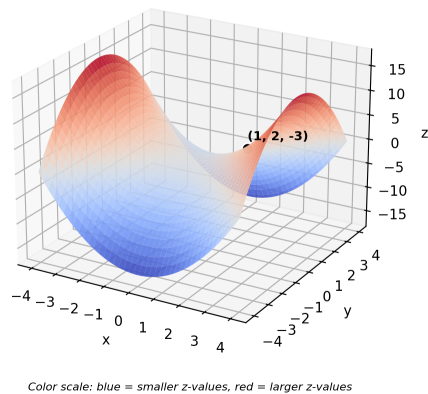


Figure (a)

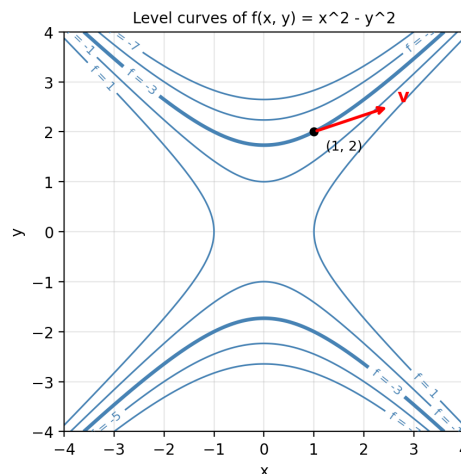


Figure (b)

- (a) Compute  $\nabla f$  at  $(1, 2)$ .
- (b) The level curve of  $f$  through  $(1, 2)$  (i.e.,  $f(x, y) = -3$ ) is shown in Figure (b), along with two neighboring level curves on either side. On the graph, draw the gradient vector  $\nabla f$  at  $(1, 2)$ , in the correct direction (you do not need to scale it precisely). Briefly explain how you knew which way it should point.
- (c) The red arrow in Figure (b) represents the direction vector  $\mathbf{v} = \langle 3, 1 \rangle$ . Compute the directional derivative of  $f$  at  $(1, 2)$  in that direction, and determine whether  $f$  is increasing or decreasing. Use the graph to confirm your conclusion.